Portfolio Optimisation
IM21204

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Theory

When constructing a portfolio, an investor allocates capital across a set of assets. The proportion of total capital assigned to each asset is called its **portfolio weight**. These weights collectively define the portfolio. They tell you which assets you hold, based on the constraints/ conditions. This assignment talks about a target return constraint namely taking the examples of 10% and 15%.. A key rule is that all weights must add up to one, since the entire capital must be deployed somewhere.

The target return constraint is an equality constraint. It requires the portfolio to deliver a specific expected return. The optimiser must find a combination of weights that hits this exact return figure while simultaneously minimising risk. This is the central idea behind Markowitz mean-variance optimisation

let there be 'n' decision variables (in this case, n stocks)

then, decision variables : $x_1, x_2, x_3 \dots x_n$ (Investment)

weightage (represented by w) : $w_1, w_2 \dots w_n$

$\sum w_i = 1$ (as explained above)

$\sum x_i = B$ & $w_i = x_i / B$

$E(R_i) = r_i$ (expected return for asset i)

Target return constraint : $\sum_{i=1}^{n} r_i w_i \geq T$ → user defined target

$w_i \geq 0 \quad \forall i = 1, 2 \dots n$

$Cov(X, Y) = E[(X - \mu_x)(Y - \mu_y)]$

$\Sigma = E[(R_i - r_i)(R_j - r_j)]$

$$\Sigma = \begin{bmatrix} Cov(R_1,R_1) & Cov(R_1,R_2) & \cdots & Cov(R_1,R_n) \\ \vdots & & & \\ Cov(R_n,R_1) & Cov(R_n,R_2) & \cdots & Cov(R_n,R_n) \end{bmatrix}$$

{ diagonal entries are $Var(R_i)$ }

finally the optimisation problem becomes :-

$$\min \ \vec{w}^T \Sigma w$$

$$\text{Such that} : \sum_{i=1}^{n} w_i = 1 \quad ; \quad \begin{pmatrix} w_1 \\ \vdots \\ w_n \end{pmatrix} \geq 0 \quad ; \quad \sum_{i=1}^{n} r_i w_i \geq T$$

Dataset

[redacted]

p=sharing

Companies selected:

Based on highest average rate of return and lowest standard deviation we have selected the following companies:

1. Titan
2. Tata Steel
3. SBIN
4. NTPC
5. Coal India
6. JSW Steel
7. Axis Bank
8. Reliance
9. HINDALCO

The expected daily and annual returns, Standard Deviation, and Analysis of the NIFTY Companies' data is mentioned in the attached sheet, 'N21 Analysis'.

Gold ETF selected:

Tata Gold ETF

What is an ETF and why have we selected it?

An ETF, or Exchange Traded Fund, is a financial instrument that trades on a stock exchange like a regular stock, but instead of representing ownership in a single company, it tracks the price of an underlying asset in this case, gold. We have selected the Tata Gold ETF rather than physical gold for one simple reason: compatibility. Portfolio optimisation requires a consistent daily price series to compute returns, variance, and covariance across all assets. The Tata Gold ETF is listed on the NSE and produces exactly that, making it directly comparable with the other stocks in our dataset. Physical gold cannot be traded or measured in this way within an equity framework.

Problem Code:

[redacted]

m1GponWRe

Optimal Portfolio allocation for the rates of return:

15% -

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[**********100%**********]  10 of 10 completed
Correlation Matrix:
              Titan  Tata Steel   SBIN   NTPC  Coal India  JSW Steel  \
Titan         1.000       0.287  0.162  0.297       0.434      0.433
Tata Steel    0.287       1.000  0.216  0.495       0.359      0.641
SBIN          0.162       0.216  1.000  0.362       0.098      0.069
NTPC          0.297       0.495  0.362  1.000       0.457      0.474
Coal India    0.434       0.359  0.098  0.457       1.000      0.486
JSW Steel     0.433       0.641  0.069  0.474       0.486      1.000
Axis Bank     0.293       0.316  0.082  0.248       0.372      0.353
Reliance      0.392       0.194  0.135  0.229       0.306      0.286
HINDALCO      0.410       0.421  0.221  0.602       0.704      0.553
Tata Gold ETF 0.136       0.169  0.041  0.153       0.206      0.178

              Axis Bank  Reliance  HINDALCO  Tata Gold ETF
Titan             0.293     0.392     0.410          0.136
Tata Steel        0.316     0.194     0.421          0.169
SBIN              0.082     0.135     0.221          0.041
NTPC              0.248     0.229     0.602          0.153
Coal India        0.372     0.306     0.704          0.206
JSW Steel         0.353     0.286     0.553          0.178
Axis Bank         1.000     0.306     0.243          0.280
Reliance          0.306     1.000     0.294          0.212
HINDALCO          0.243     0.294     1.000          0.127
Tata Gold ETF     0.280     0.212     0.127          1.000

Covariance Matrix (Σ) constructed from manual std devs + historical correlations.
 OPTIMAL PORTFOLIO WEIGHTS
Asset                     Weight    Allocation %
------------------------------------------------
Titan                     0.0262            2.62%
Tata Steel                0.0000            0.00%
SBIN                      0.0781            7.81%
NTPC                      0.2029           20.29%
Coal India                0.0000            0.00%
JSW Steel                 0.0887            8.87%
Axis Bank                 0.1781           17.81%
Reliance                  0.4259           42.59%
HINDALCO                  0.0000            0.00%
Tata Gold ETF             0.0000            0.00%
------------------------------------------------
Sum of Weights            1.0000

Portfolio Expected Return : 15.00%
Portfolio Std Deviation   : 14.43%
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10% -

```
[**********100%**********]  10 of 10 completed

Correlation Matrix:
              Titan  Tata Steel   SBIN   NTPC  Coal India  JSW Steel  \
Titan         1.000       0.287  0.162  0.297       0.434      0.433
Tata Steel    0.287       1.000  0.216  0.495       0.359      0.641
SBIN          0.162       0.216  1.000  0.362       0.098      0.069
NTPC          0.297       0.495  0.362  1.000       0.457      0.474
Coal India    0.434       0.359  0.098  0.457       1.000      0.486
JSW Steel     0.433       0.641  0.069  0.474       0.486      1.000
Axis Bank     0.293       0.316  0.082  0.248       0.372      0.353
Reliance      0.392       0.194  0.135  0.229       0.306      0.286
HINDALCO      0.410       0.421  0.221  0.602       0.704      0.553
Tata Gold ETF 0.136       0.169  0.041  0.153       0.206      0.178

              Axis Bank  Reliance  HINDALCO  Tata Gold ETF
Titan             0.293     0.392     0.410          0.136
Tata Steel        0.316     0.194     0.421          0.169
SBIN              0.082     0.135     0.221          0.041
NTPC              0.248     0.229     0.602          0.153
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JSW Steel         0.353     0.286     0.553          0.178
Axis Bank         1.000     0.306     0.243          0.280
Reliance          0.306     1.000     0.294          0.212
HINDALCO          0.243     0.294     1.000          0.127
Tata Gold ETF     0.280     0.212     0.127          1.000

Covariance Matrix (Σ) constructed from manual std devs + historical correlations.

  OPTIMAL PORTFOLIO WEIGHTS
 Asset                    Weight    Allocation %
 ------------------------------------------------
 Titan                    0.0000           0.00%
 Tata Steel               0.0000           0.00%
 SBIN                     0.0000           0.00%
 NTPC                     0.1963          19.63%
 Coal India               0.0000           0.00%
 JSW Steel                0.1081          10.81%
 Axis Bank                0.1915          19.15%
 Reliance                 0.5041          50.41%
 HINDALCO                 0.0000           0.00%
 Tata Gold ETF            0.0000           0.00%
 ------------------------------------------------
 Sum of Weights           1.0000

 Portfolio Expected Return : 10.00%
 Portfolio Std Deviation   : 15.41%
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Portfolio Growth Pattern:

